

ARC322

Networking lens on resiliency architectures

John Formento

Principal Solutions Architect
Amazon Web Services

Tarik Makota

Sr. Principal Solutions Architect
Amazon Web Services



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.

Agenda

Common network availability questions for:

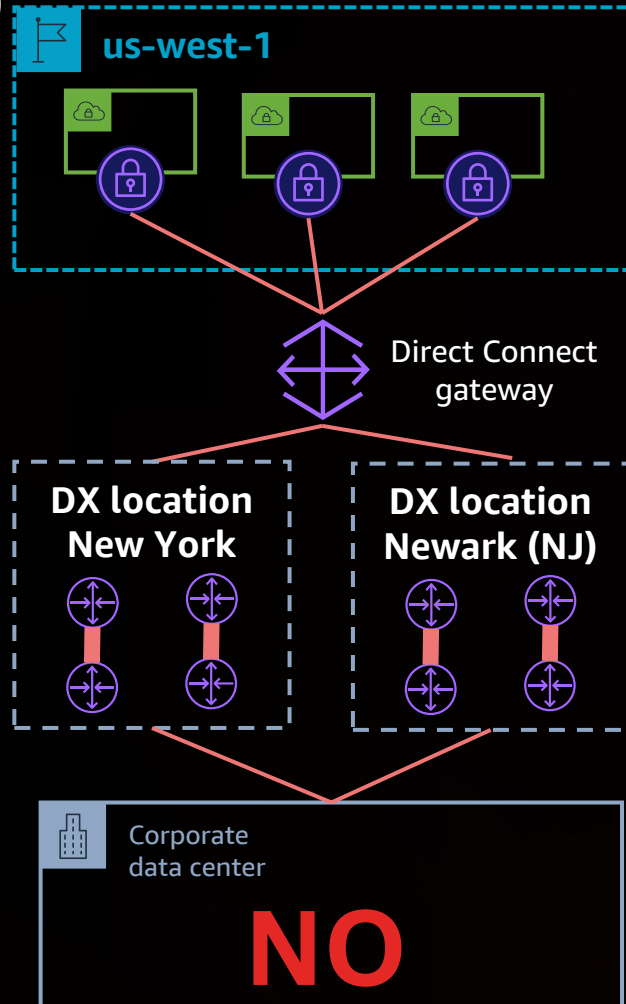
- Hybrid
- Connectivity on AWS
- Load balancing

Hybrid

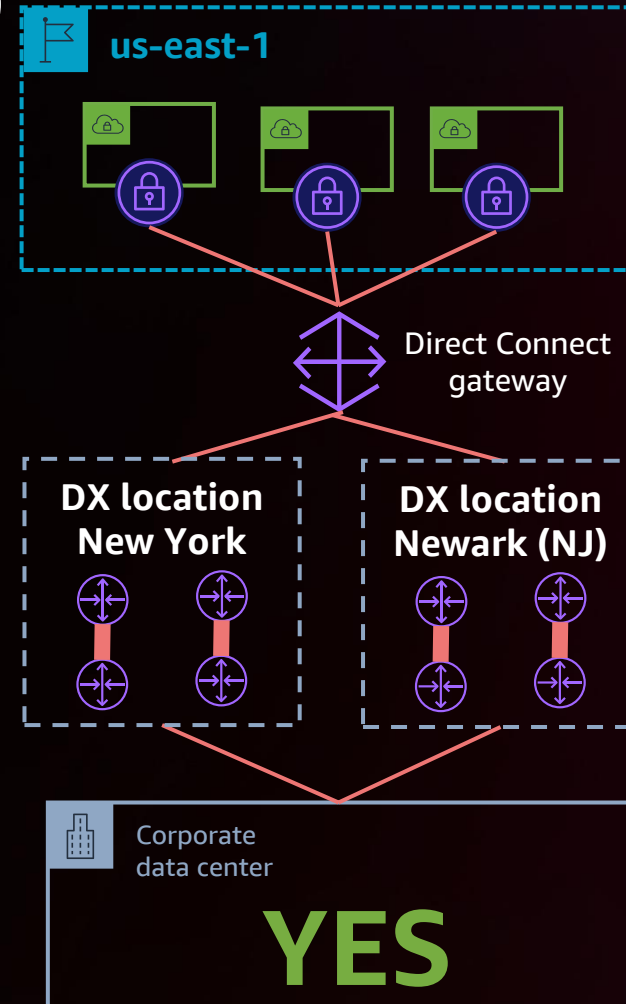


How to get 99.99% SLA on AWS Direct Connect?

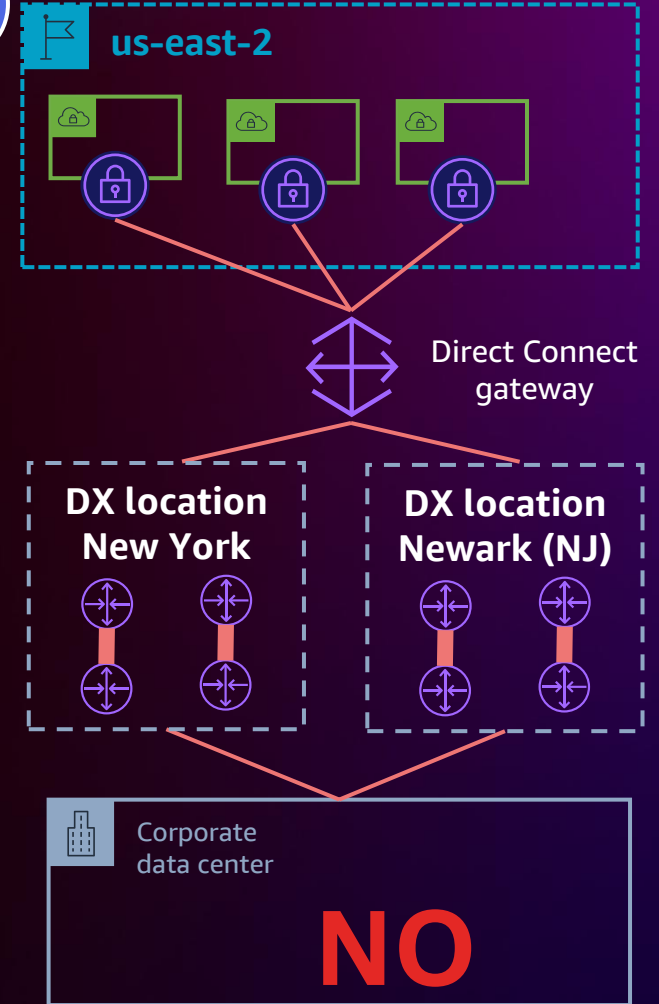
1



2



3



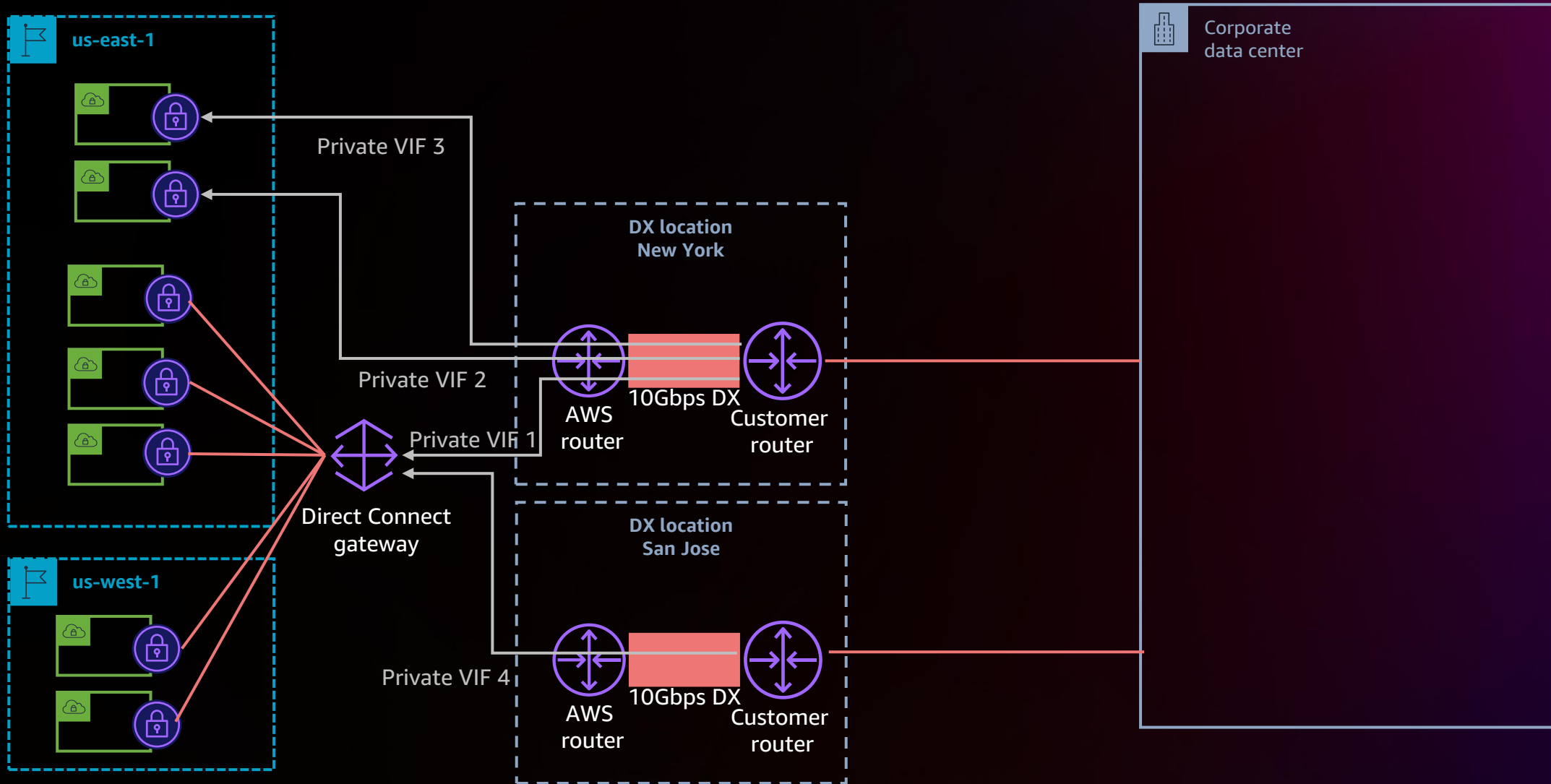
How to get 99.99% SLA on AWS Direct Connect?

99.99%

- Included Resource uses virtual interfaces on at least 4 Dedicated Connections across a minimum of 2 Direct Connect locations (with no fewer than 2 connections in a single location), and at least one of the Direct Connect locations uses the Associated AWS Region in which your workload is located.
- Each Dedicated Connection must terminate on a unique AWS endpoint as indicated by the AWS Device ID visible in the AWS Console and Direct Connect APIs.
- Each virtual interface provisioned on the Included Resource must be configured for full, independent access to your workload.
- You must have the [Enterprise Support](#) plan in place and complete a “Well-Architected Review” with an AWS Solutions Architect.
- For private endpoints, your workload must be deployed to two or more Availability Zones.
- You must request your AWS Solutions Architect to provide AWS Enterprise Support with a list of the Included Resource IDs (e.g., AWS Virtual Private Gateway and its associated virtual interface IDs) that meet the Minimum Configuration Requirements to qualify for Service Credits.



Resilience improves if I add a backup DXGW?

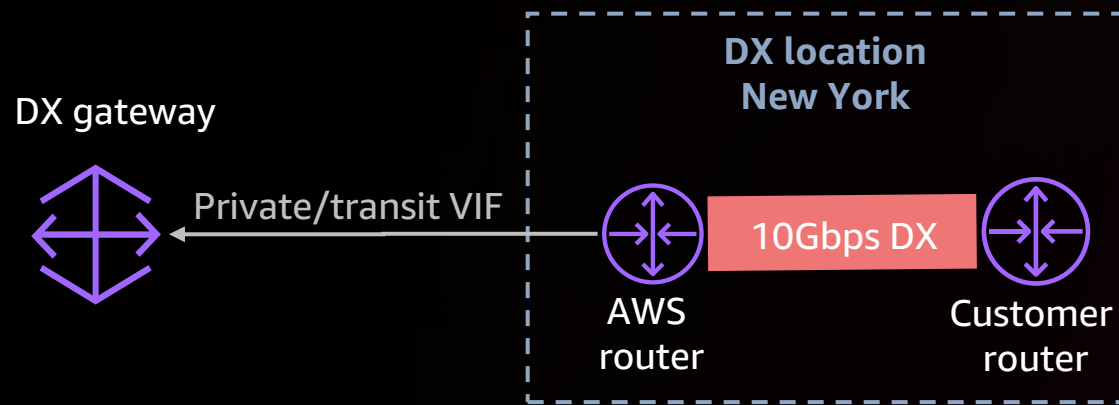


Resilience improves if I add a backup DXGW?

NO

- DXGW is **not** a single point of failure
- DXGW is a **configuration deployed on existing network devices**, therefore it doesn't change performance or availability
- DXGW **expands connectivity options for** a DX location to multiple Regions, making creating resilient setups easier

How long will it take for DX to detect a link failure on a virtual interface (VIF)?



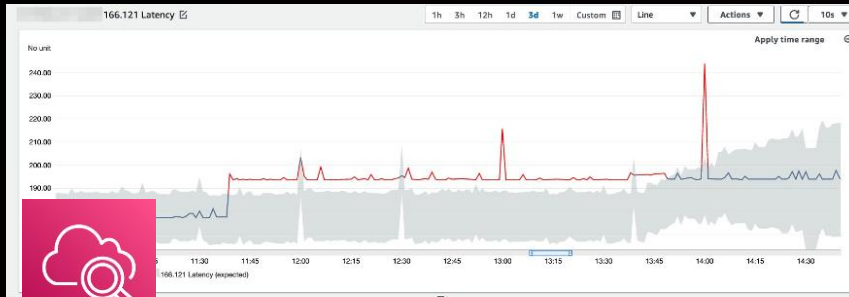
eBGP will detect a failed peer in up to 90 seconds (BGP timers)

Bidirectional Forwarding Detection (BFD) will detect a failed peer in less than 1 second (3x 300ms)

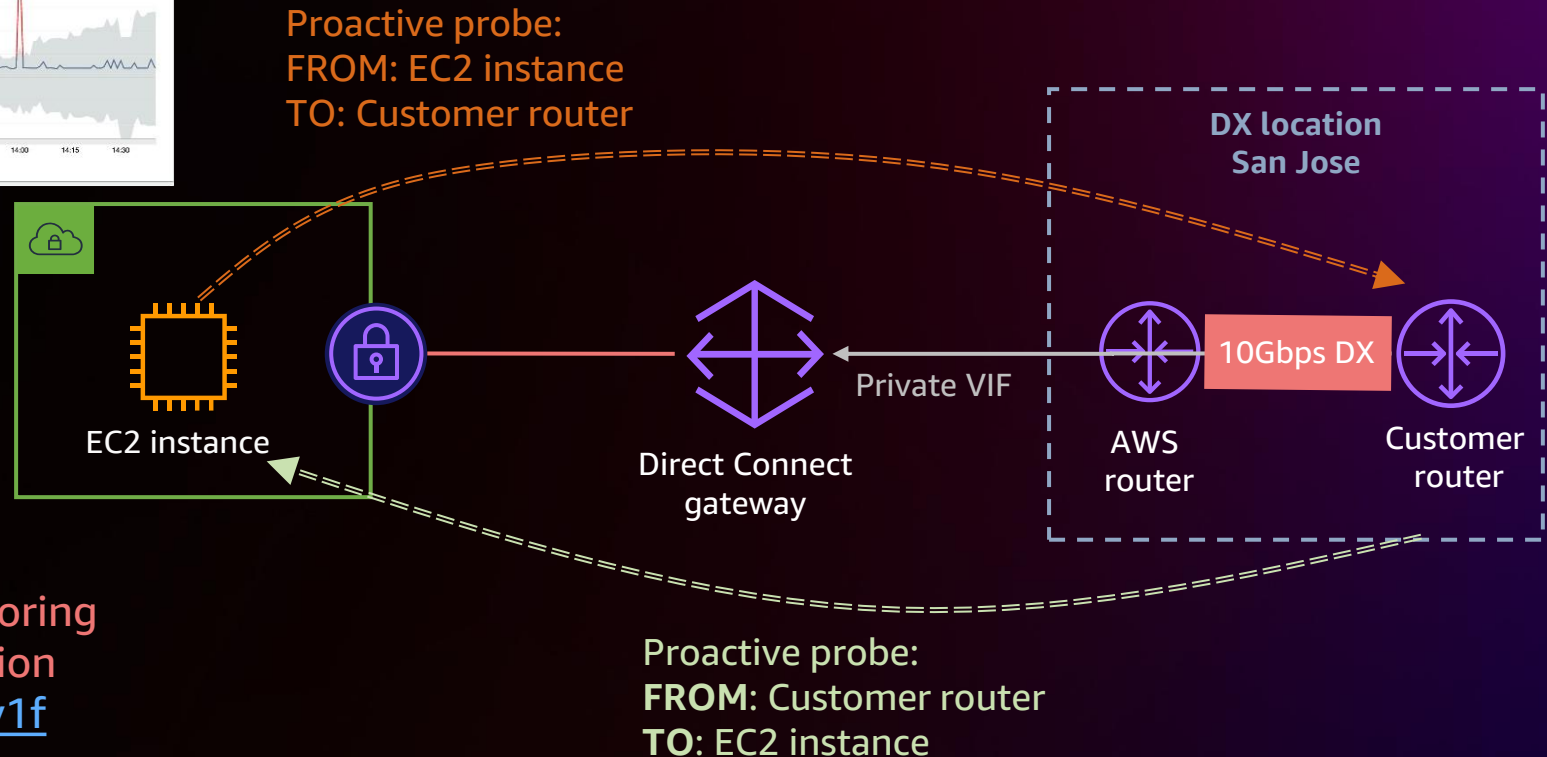


Does DX support monitoring all the way to VPC resource?

NOT natively, but workarounds are available



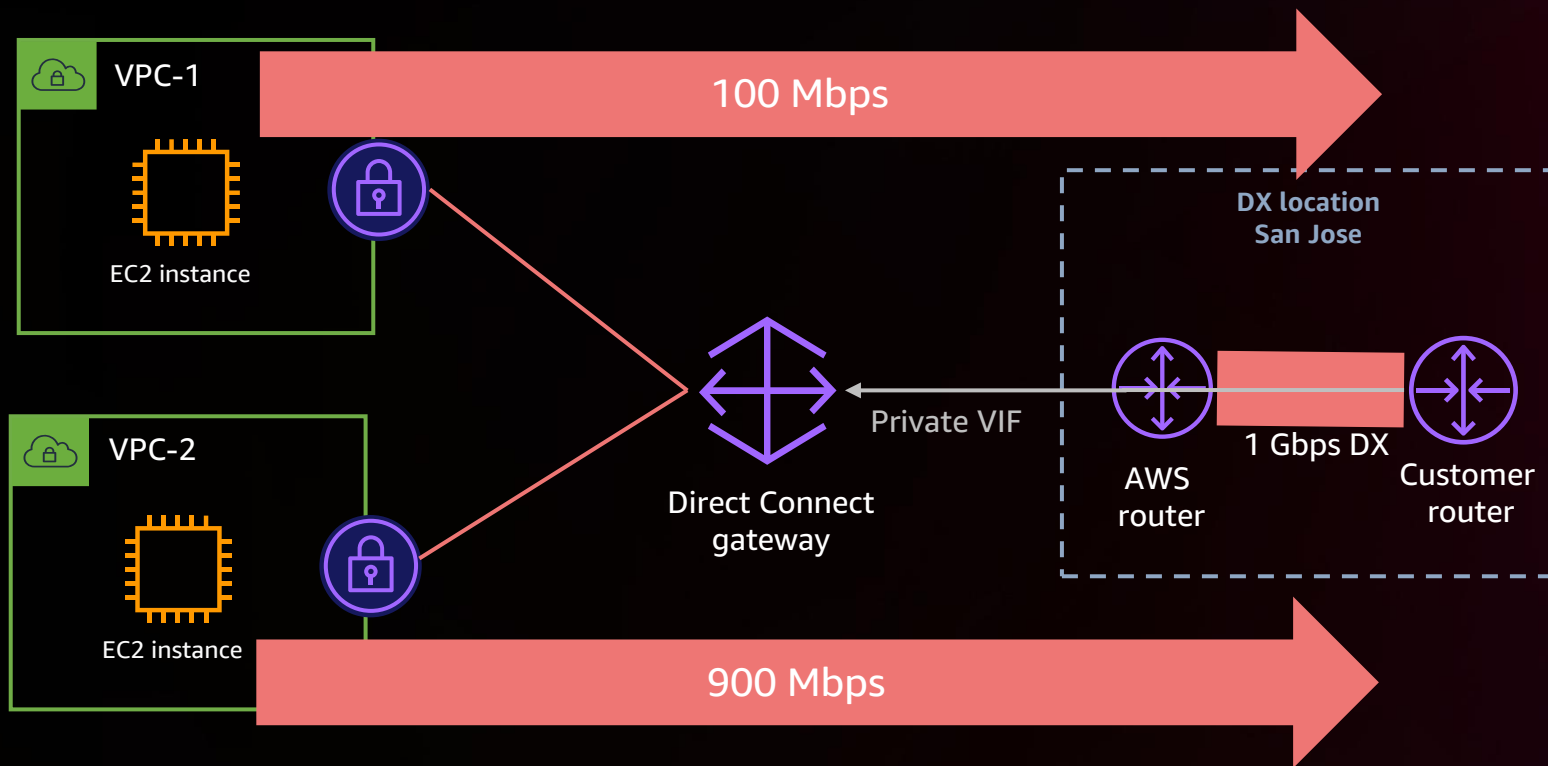
Amazon CloudWatch



BLOG: AWS Direct Connect monitoring and failover with Anomaly Detection
<https://tiny.amazon.com/190i4yy1f>

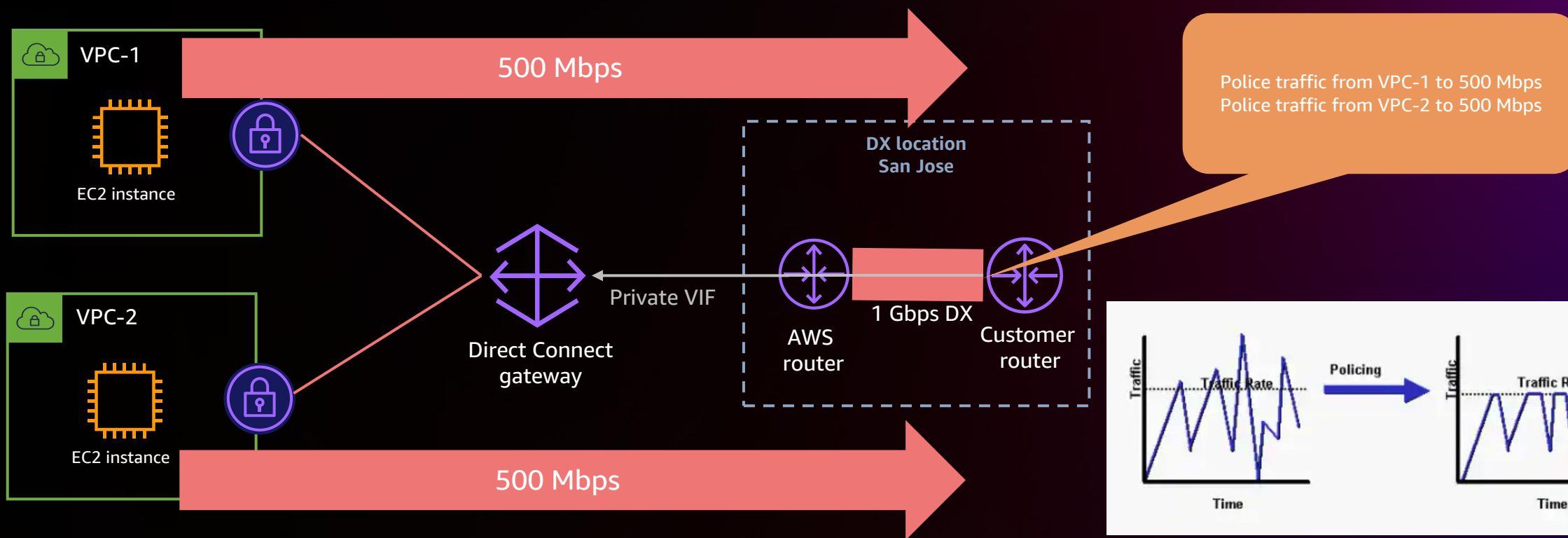
Can I police traffic on DX to protect from noisy neighbors?

NOT natively, but workarounds are available



Can I police traffic on DX to protect from noisy neighbors?

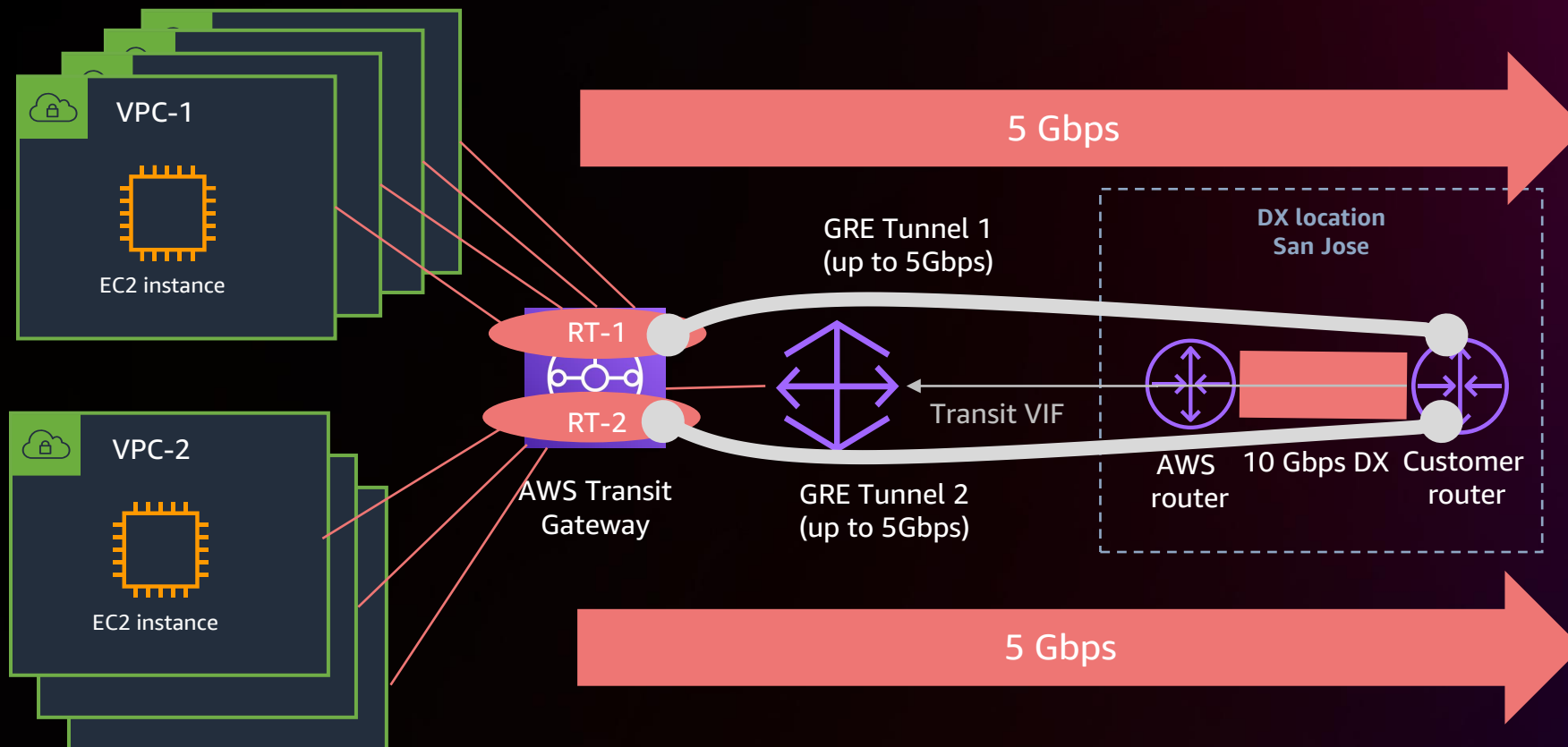
NOT natively, but workarounds are available



* Policing on customer router will not protect DX from getting congested by UDP flows originating from AWS

Can I police traffic on DX to protect from noisy neighbors?

NOT natively, but workarounds are available

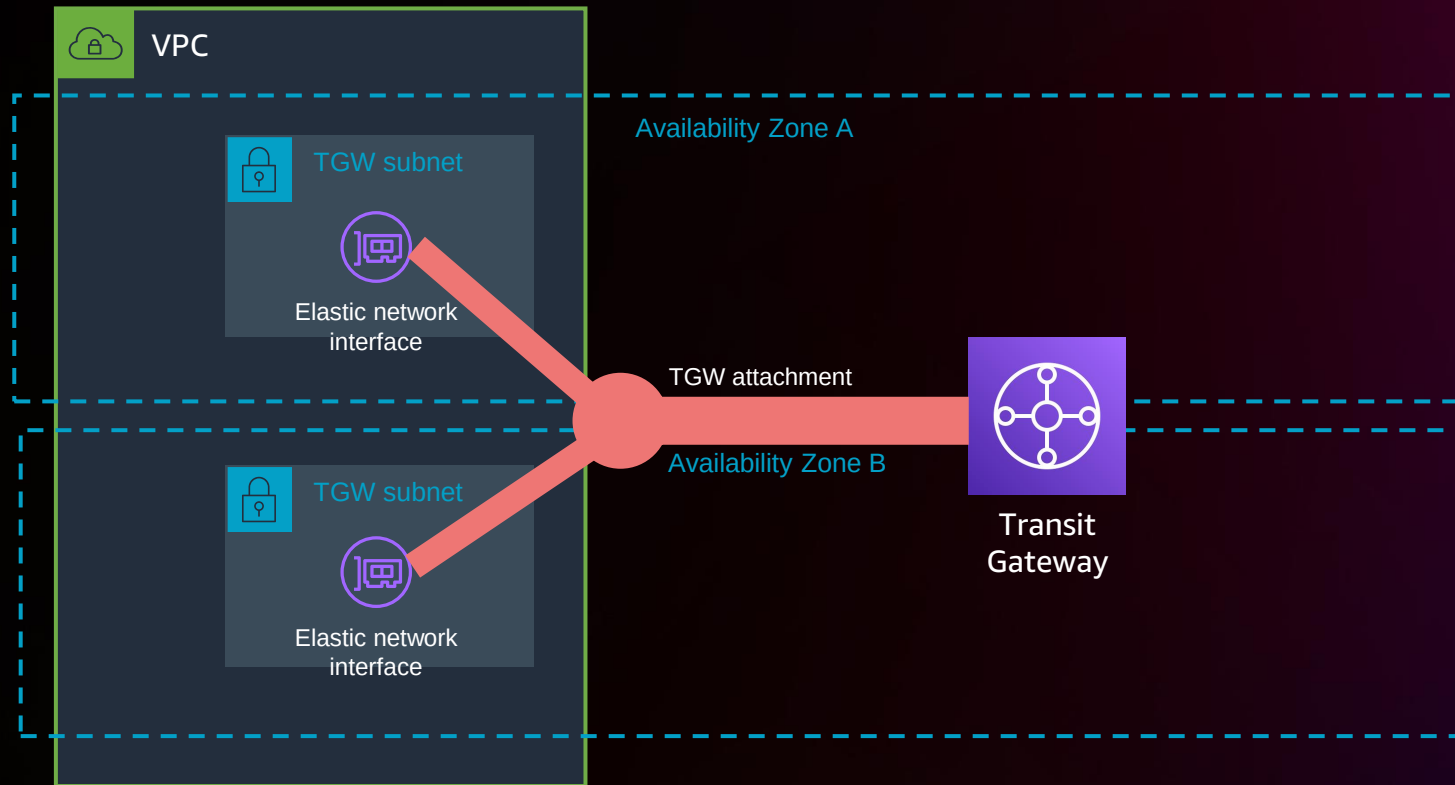


Connectivity on AWS



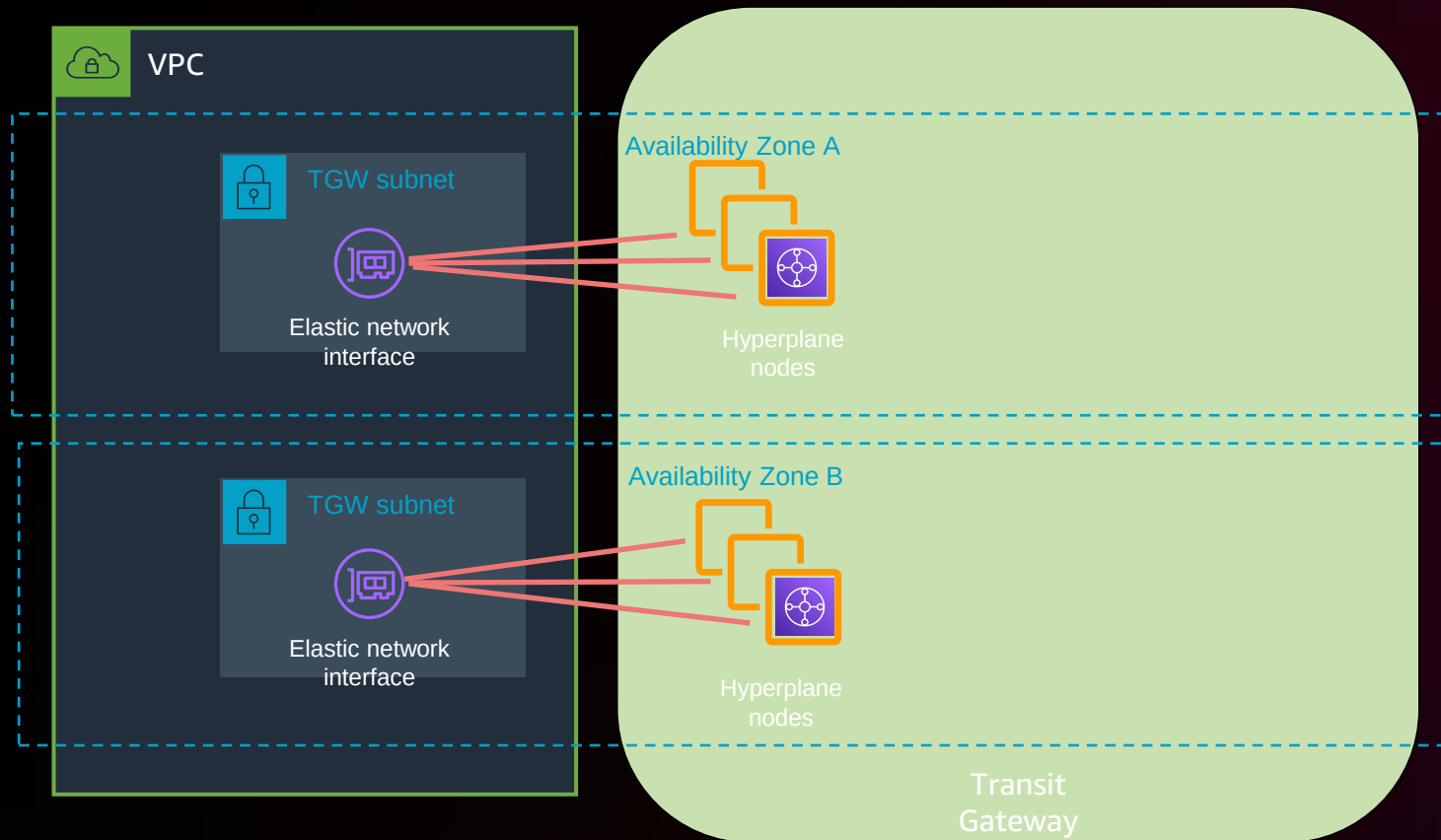
Resilience improves if I add a backup TGW?

NOT for data plane



Resilience improves if I add a backup TGW?

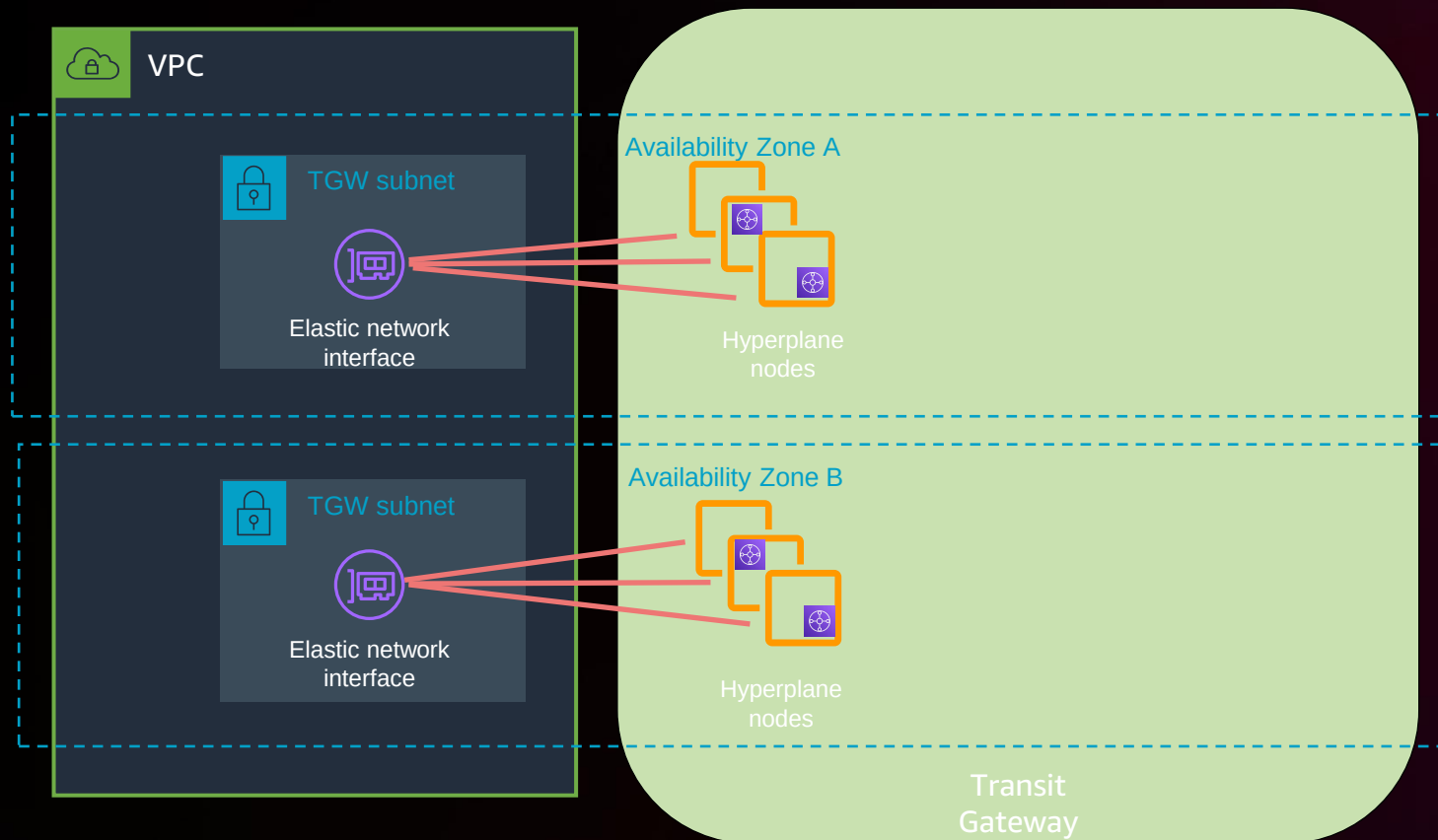
Not for data plane



TGW SLA: 99.99%

Resilience improves if I add a backup TGW?

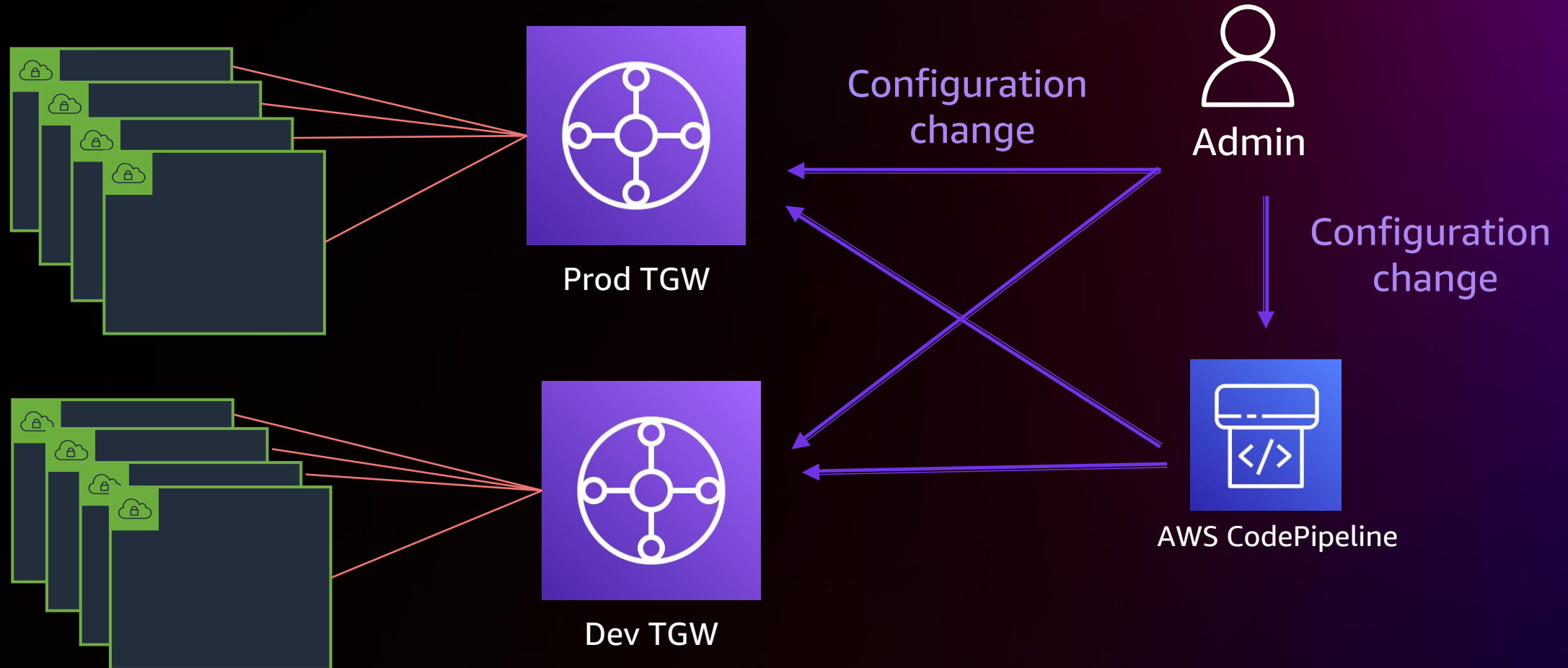
Not for data plane



TGW SLA: 99.99%

Resilience improves if I add a backup TGW?

Not for data plane. Could be an option for management plane

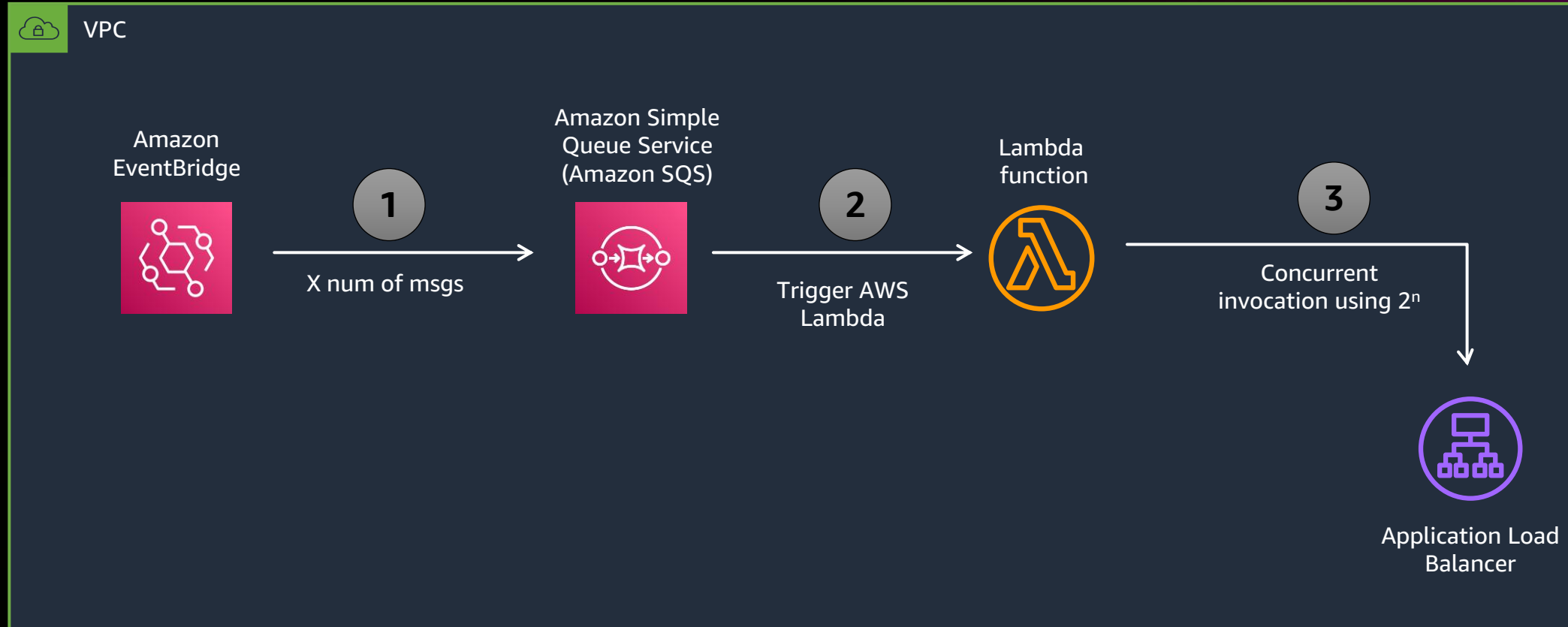


Load balancing



Traffic can increase 10x almost instantly – Can I prewarm my ALB by myself?

YES, you can



Similar solution: <https://tiny.amazon.com/op4kgz6d>

Credit: Luis Sena

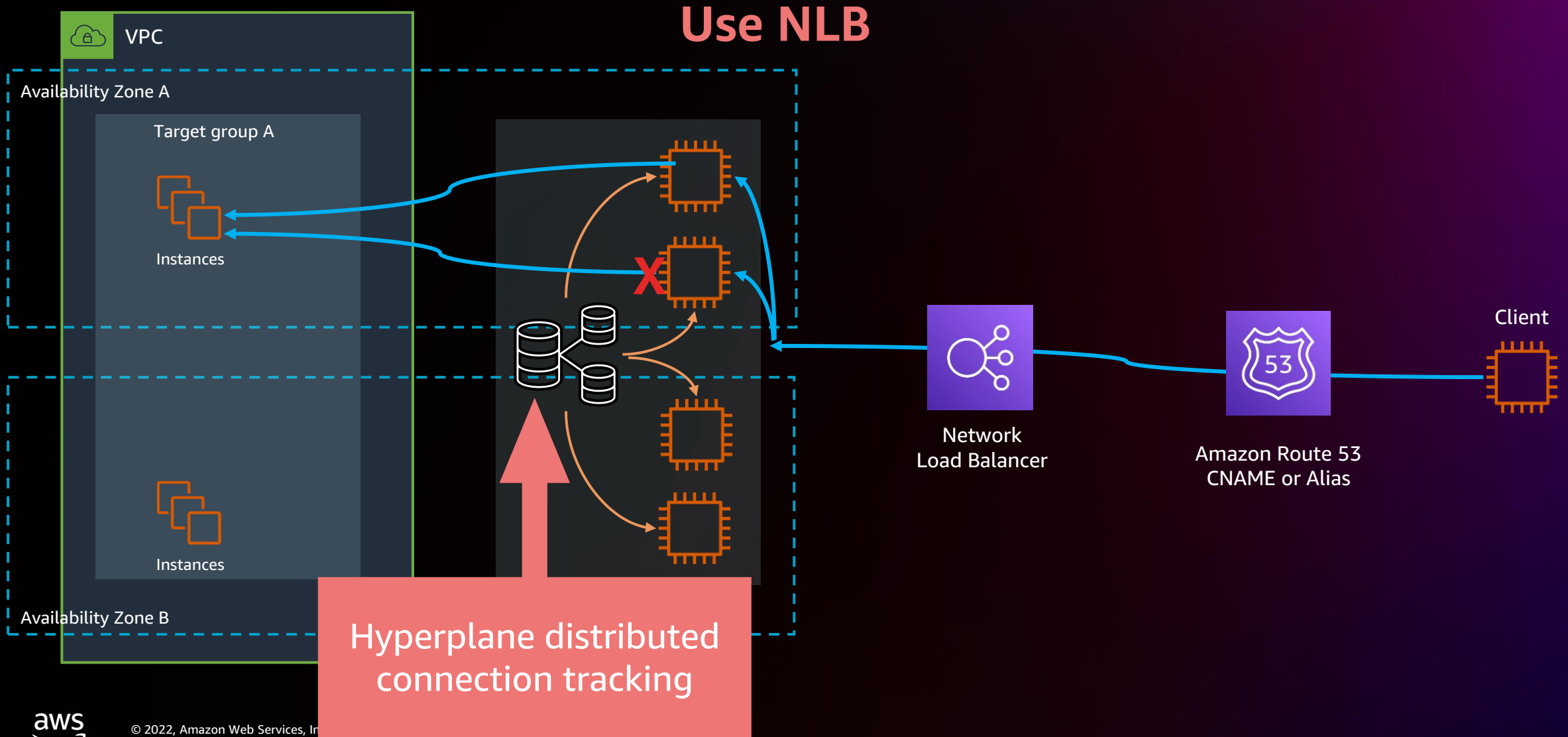


© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.



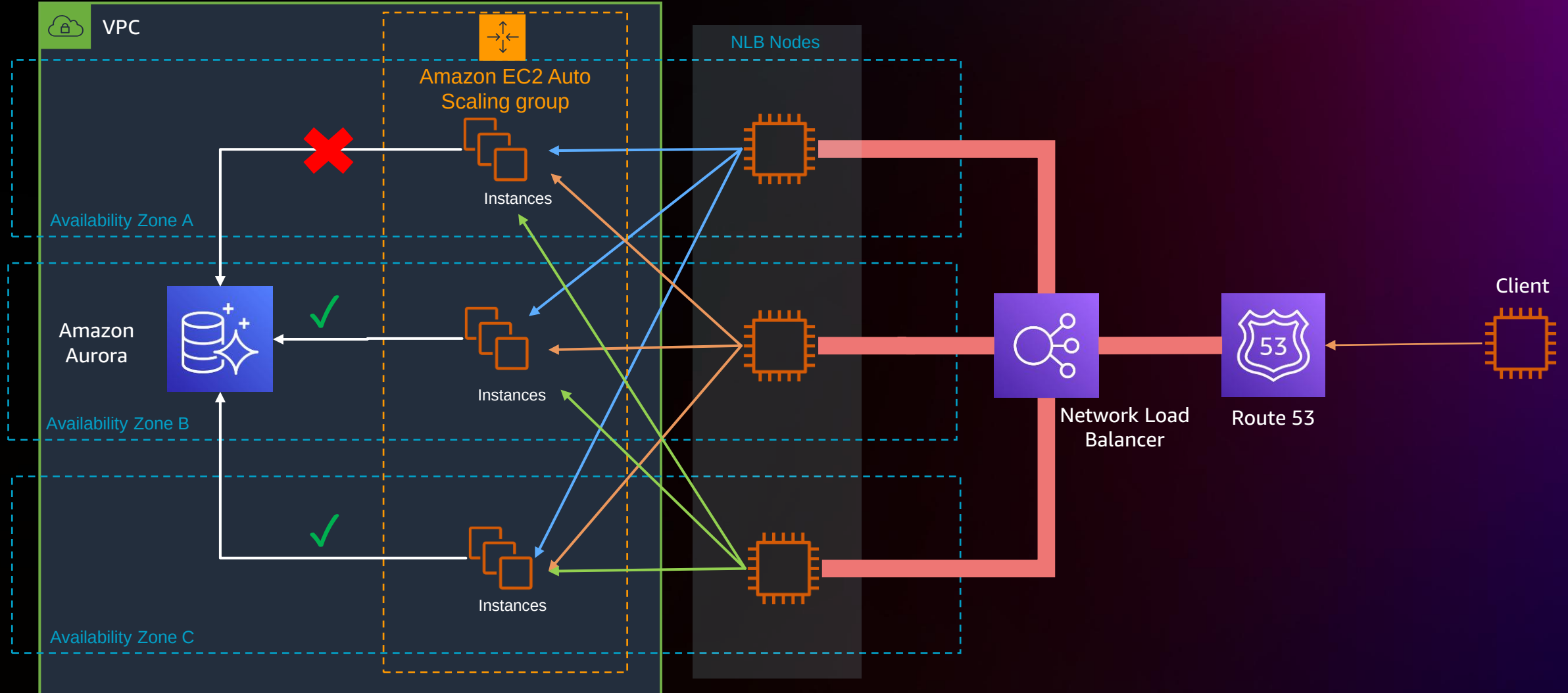
SCAN ME

Which load balancer to use for long-lived TCP flows?



Should I enable cross-zone load balancing for NLB?

When it comes to resilience, probably NO



Thank you!

John Formento

formjohn@amazon.com

Tarik Makota

tmak@amazon.com



Please complete the session survey in the **mobile app**



© 2022, Amazon Web Services, Inc. or its affiliates. All rights reserved.